Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*A technology enabling high-pressure PEM Electrolysis development by degradation diagnostics*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	New/renewable energy	Hydrogen - manufacturing	RnD / Engineering Services	Testing / Inspection / Analysis Techniques
Concept of Proposed Technology	material characterization platform; multi-modal Identification of critical degradation mechanisms for safety of long-term high-pressure operation			
Rationale for Joint R&D Project	Collaboration combines Fraunhofer's characterization expertise with Korean SMEs' quality control testing and manufacturing of pressurized PEM-electrolysers			
Key Words	1. Defect Diagnostics for PEM 2. Cross-scale analysis methods 3. Spatially resolved analysis			

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	• Root cause analysis of defects and degradation in high pressure PEM electrolysis • Understanding the influence of high-pressure PEM operation on the materials, especially the pressure-driven deformation and degradation of membranes and gaskets • Identification of critical degradation mechanisms for safety of long-term high-pressure operation			
R&D Target Specifications	- Cross-scale, spatially resolved analysis of PEM MEAs operating in high-pressure PEM - Identification of critical components and failure modes - Correlation between manufacturing-induced heterogeneities and operational failure - Development of stress-tests for the evaluation of materials - Development of new diagnostics methods and workflows for existing methods for the evaluation of materials - Lifetime prediction of materials			

R&D Plan

Phase	TRL goal	Research approach & methodologies	Key tasks
1	3 (lab-scale verification)	Bench-scale diagnostic set-up; materials/component screening; baseline pressure-impact studies	Prepare instrumented lab cells and fixtures for controlled pressurization. - Collect seed datasets; establish data quality gates and analysis protocols. - Map early failure signatures vs. material heterogeneities.
2	4 (key performance evaluation)	Cross-scale and spatially resolved analysis; accelerated stress testing; workflow development	- Execute pressure cycling, humidity and temperature stress tests. - Develop diagnostic workflows (e.g., mechanical, electrochemical, and microstructural routines). - Quantify deformation and degradation pathways in membranes and gaskets. - Correlate manufacturing heterogeneities with performance metrics.
3	5 (prototype production and evaluation)	Prototype high-pressure cells/short stacks; integrated diagnostics; safety and reliability checks	- Manufacture prototype cells and, if feasible, short stacks. - Integrate monitoring and diagnostic routines into operation. - Evaluate performance, safety indicators, and degradation rates under realistic profiles.
4	6 (pilot-scale validation)	Pilot demonstration in production-like environment; reliability validation	- Run pilot-scale tests with agreed operating envelopes. - Validate reliability, repeatability, and scale-up readiness. - Document protocols, training materials, and transfer package.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Degradation diagnostics on stressed cells and materials • Development of measurement routines • Ex-situ stress testing development • Identification of critical processes leading to material degradation
Expected R&R of Korean SME Partner	• Manufacturing of high-pressure PEM cells • Cell testing and if necessary short-stack testing • Electrochemical analysis of cells during operation • Cell commercializing

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	• Diagnostics framework and understanding about most critical material processes • Operationability indicators for high-pressure PEM Electrolysis • Risk reduction for the scale-up of high-pressure stack development

6. References

(1)

(2)

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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